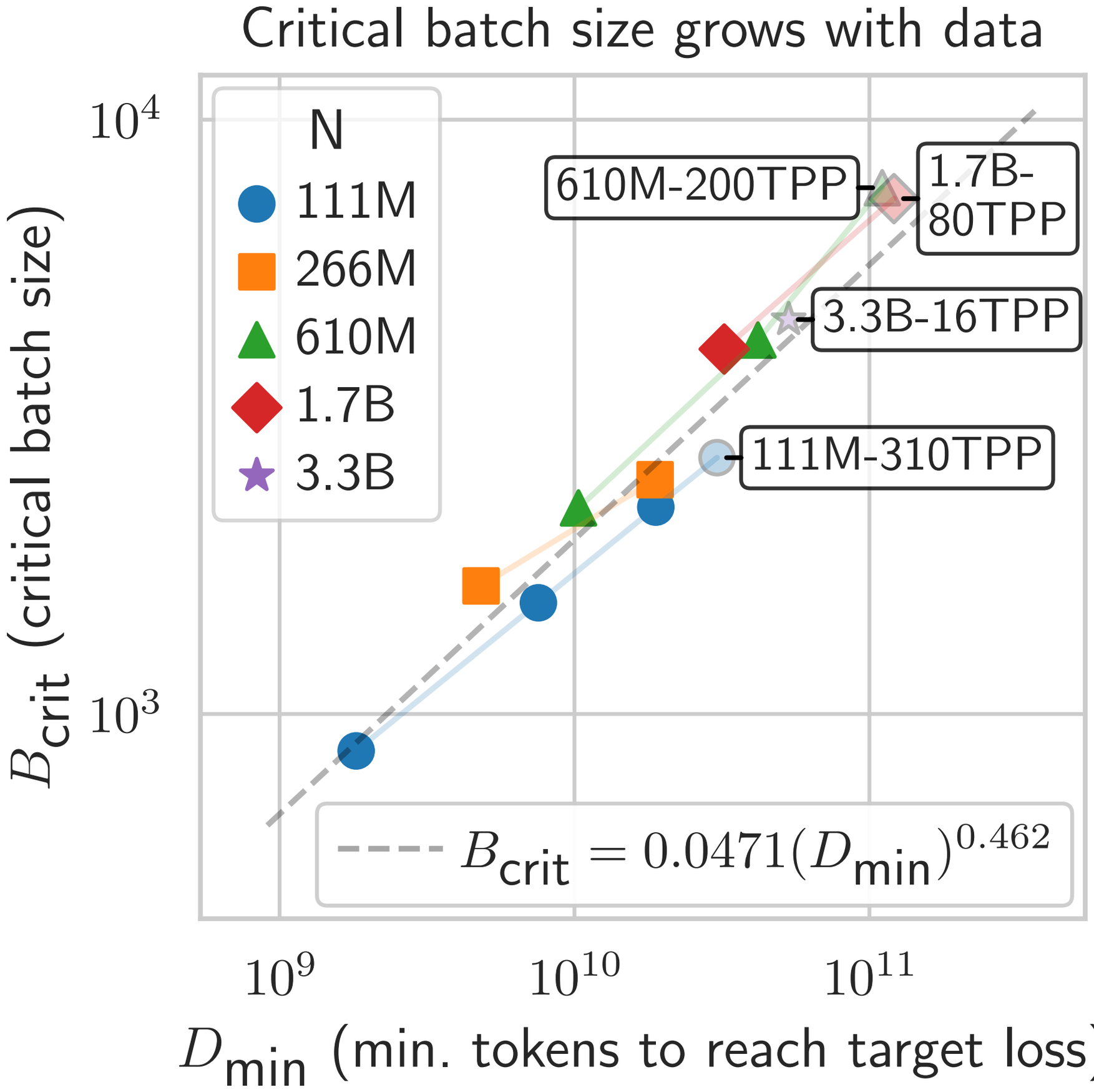
Critical batch size grows with data
N
111M
266M
610M
1.7B
3.3B
610M-200TPP
1.7B-80TPP
3.3B-16TPP
111M-310TPP
$B_{\text{crit}} = 0.0471(D_{\text{min}})^{0.462}$
10^4
10^3
10^9
10^{10}
10^{11}
B_{crit} (critical batch size)
D_{min} (min. tokens to reach target loss)